

CH 224 – Organic Chemistry II Lab Syllabus

1 Credit Hour

Course Description

Laboratory experience to accompany CH 223. Introduction to basic organic laboratory equipment and techniques.

Learning Outcomes

Upon successful completion of this course, the student will be able to:

1. Become aware of the safety procedures and regulations that must be observed when performing organic chemistry experiments in a laboratory.
2. Use the techniques learned in CH222 in order to perform advanced synthesis of organic materials containing carbonyls:
 - a. Perform simple separations using Thin Layer Chromatography.
 - b. Separate complex mixtures using Column Chromatography.
 - c. Purify organic compounds using recrystallization and sublimation.
 - d. Determine melting points and boiling points and use them to characterize organic compounds.
 - e. Perform Infrared Spectroscopy and use the results obtained to identify organic compounds.
 - f. Combine the use of different synthetic and separation methods in order to execute multi-step synthesis.
3. Determine the structure of organic compounds using information gathered from their NMR spectra.
4. Perform literature searches using approved search engines and chemistry databases.
5. Complete oxidation and reduction reactions using ketones as substrates.
6. Complete electrophilic aromatic substitution reactions.
7. Perform aldol condensation reactions.
8. Complete amide preparation reactions.

9. Prepare arenediazonium salts.

10. Perform a saponification reaction on a selected triglyceride.

Course Structure and Schedule

Your lab meets on the date, room, and time indicated in your class schedule.

<u>Day</u>	<u>Experiment</u>
Tues. Jan 17 - Fri. Jan 20	Introduction to the Lab and Exp 1: Literature Search Assignment (due next lab meeting)
Tues. Jan 24 - Fri. Jan 27	Exp 2: NMR Spectroscopy
Tues. Jan 31 - Fri. Feb 3	Exp 3: Reduction of a Ketone
Tues. Feb 7 - Fri. Feb 10	Exp 4: Oxidation of Acetophenone
Tues. Feb 14 - Fri. Feb 17	NO LABS: Wellness
Tues. Feb 21 - Fri. Feb 24	LABORATORY QUIZ 1 Exp 5: Nitration of Methyl Benzoate
Tues. Feb 28 - Fri. March 3	Exp 6: Aldol Condensation
Tues. March 7 - Fri. March 10	Exp 7: Preparation of 1,4-diphenyl-1,3-Butadiene
Tues. March 14 - Fri. March 17	NO LAB - Spring Break
Tues. March 21 - Fri. March 24	Exp 8: Amide Preparation
Tues. March 28 - Fri. March 31	Exp 9: Arenediazonium Salts
Tues. April 4 - Fri. April 7	Exp 10: Hydrolysis of Glycerol Tristearate LABORATORY QUIZ 2

Teaching Assistant: Your TA is listed on Moodle. TAs will provide students with their contact information and office hours.

Dr. Cassie Lilly– Lab Director

Email: cassie_lilly@ncsu.edu

Office Location: FOX 254

Office Hours: Email for appointment

Course Materials

Textbooks

All course materials are on WebAssign and Moodle. The WebAssign Class KEY and directions for how to purchase and access the lab manual and assignments for each lab is posted on the Moodle section. No other textbooks or notebooks are required.

Requisites and Restrictions

Prerequisites: CH 221 and CH 222

Co-requisites: CH 223

Grading

Pre=labs are to be completed on WebAssign and are due 1 hour before the start of the lab period. Lab worksheets are to be completed during lab and are due at the end of each lab period.

Pre-Labs 1 and 2	10 points
All other Labs	60 points
Quiz	100 points
Worksheet 1 and 2	90 points
All other worksheets	40 points

The 10 experiment lab grades and the 2 quiz grades will be averaged together to calculate the final grade.

Letter Grades

This Course uses Standard NCSU Letter Grading:

97 ≤ **A+** ≤ 100

93 ≤ **A** < 97

90 ≤ **A-** < 93

87 ≤ **B+** < 90

83 ≤ **B** < 87

80 ≤ **B-** < 83

77	≤	C+	<	80
73	≤	C	<	77
70	≤	C-	<	73
67	≤	D+	<	70
63	≤	D	<	67
60	≤	D-	<	63
0	≤	F	<	60

Attendance Policy

Documentation is required to verify an absence. Please send documentation for excused absences to your TA. The Division of Academic and Student Affairs can assist students with documentation (<https://dasa.ncsu.edu/support-and-advocacy/find-help/absence-verification/>) If excused, you may make up the lab in person during a different section that is approved by the TA or Lab Director (cassie_lilly@ncsu.edu). It may be that a different make up opportunity is provided. It is important to notify the TA or lab director as soon as possible with the excused absence in order to coordinate an in person lab make up. More than one absence for an in person lab, even if excused, could result in an incomplete (I) or failure of the course (F). This will apply if the first excused absence was not made up in person during a different lab section.

An unexcused absence for one lab will result in a score of 0 for that lab. Missing two or more labs for unexcused reasons will result in an F.

For complete attendance and excused absence policies, please see <http://policies.ncsu.edu/regulation/reg-02-20-03>

Requirements for Credit-Only (S/U) Grading

In order to receive a grade of S, students are required to take all exams and quizzes, complete all assignments, and earn a grade of C- or better. Conversion from letter grading to credit only (S/U) grading is subject to university deadlines. Refer to the Registration and Records calendar for deadlines related to grading. For more details refer to <http://policies.ncsu.edu/regulation/reg-02-20-15>.

Requirements for Auditors (AU)

Information about and requirements for auditing a course can be found at <http://policies.ncsu.edu/regulation/reg-02-20-04>.

Policies on Incomplete Grades

If an extended deadline is not authorized by the instructor or department, an unfinished incomplete grade will automatically change to an F after either (a) the end of the next regular semester in which the student is enrolled (not including summer sessions), or (b) the end of 12 months if the student is not enrolled, whichever is shorter. Incompletes that change to F will count as an attempted course on transcripts. The burden of fulfilling an incomplete grade is the responsibility of the student. The university policy on incomplete grades is located at <http://policies.ncsu.edu/regulation/reg-02-50-3>.

Incomplete grades will only be granted if more than 80% of the total course requirements have been fulfilled.

Late Assignments

If you need an extension for an assignment, contact your TA.

If you contact your TA during the week after the due date, you will receive a no-penalty extension.

If you contact your TA more than a week after the due date, you will receive an extension with a 50% penalty.

Academic Integrity

Academic Integrity

Students are required to comply with the university policy on academic integrity found in the Code of Student Conduct found at <http://policies.ncsu.edu/policy/pol-11-35-01>

None.

Academic Honesty

See <http://policies.ncsu.edu/policy/pol-11-35-01> for a detailed explanation of academic honesty.

None.

Honor Pledge

Your signature on any test or assignment indicates "I have neither given nor received unauthorized aid on this test or assignment."

Electronically-Hosted Course Components

Students may be required to disclose personally identifiable information to other students in the course, via electronic tools like email or web-postings, where relevant to the course. Examples

include online discussions of class topics, and posting of student coursework. All students are expected to respect the privacy of each other by not sharing or using such information outside the course.

Accommodations for Disabilities

Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, student must register with the Disability Services Office (<http://www.ncsu.edu/dso>), 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation at <http://policies.ncsu.edu/regulation/reg-02-20-01>.

Non-Discrimination Policy

NC State University provides equality of opportunity in education and employment for all students and employees. Accordingly, NC State affirms its commitment to maintain a work environment for all employees and an academic environment for all students that is free from all forms of discrimination. Discrimination based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation is a violation of state and federal law and/or NC State University policy and will not be tolerated. Harassment of any person (either in the form of quid pro quo or creation of a hostile environment) based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation also is a violation of state and federal law and/or NC State University policy and will not be tolerated. Retaliation against any person who complains about discrimination is also prohibited. NC State's policies and regulations covering discrimination, harassment, and retaliation may be accessed at <http://policies.ncsu.edu/policy/pol-04-25-05> or http://www.ncsu.edu/equal_op/. Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 919-515-3148.

Course Schedule: All labs are due at 11:59pm on the dates shown next to the experiment number